

## Assignment 3

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### Problem 1

$$a) 1) H(s) = \frac{V_o}{V_i}$$

$$V_o = \frac{V_i \cdot R}{\frac{1}{sC} + R} \Rightarrow \frac{V_o}{V_i} = \frac{R}{\frac{1}{sC} + R} = \frac{RCs}{1 + RCs}$$

$$H(j\omega) = \frac{RCj\omega}{1 + RCj\omega}$$

$$2) V_o = \frac{V_i \cdot R}{sL + R} \Rightarrow \frac{V_o}{V_i} = \frac{R}{sL + R}$$

$$H(j\omega) = \frac{R}{Lj\omega + R}$$

$$3) \frac{V_o}{V_i} = \frac{\frac{1}{sC}}{\frac{1}{sC} + R} = \frac{1}{RCs + 1}$$

$$H(j\omega) = \frac{1}{RCj\omega + 1}$$

$$4) V_o = \frac{V_i \cdot sL}{R + sL} \Rightarrow \frac{V_o}{V_i} = \frac{sL}{R + sL}$$

$$H(j\omega) = \frac{j\omega L}{R + j\omega L}$$

$$b) 1) |H(j\omega)| = \frac{|RCj\omega|}{|1 + RCj\omega|} = \frac{RC\omega}{\sqrt{(RC\omega)^2 + 1}}$$

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}} |H|_{\max} \Rightarrow |H(j\omega_c)| = \frac{1}{\sqrt{2}}$$

$$\frac{RC\omega_c}{\sqrt{(RC\omega_c)^2 + 1}} = \frac{1}{\sqrt{2}}$$

$$R^2 C^2 \omega_c^2 = \frac{1}{2} R^2 C^2 \omega_c^2 + \frac{1}{2}$$

$$\omega_c (R^2 C^2 - \frac{1}{2} R^2 C^2) = \frac{1}{2}$$

$$\omega_c = \frac{\frac{1}{2}}{\frac{RC}{2}} = \frac{1}{RC}$$

$$2) |H(j\omega)| = \frac{R}{\sqrt{(L\omega)^2 + R^2}}$$

$$\frac{R}{\sqrt{(L\omega_c)^2 + R^2}} = \frac{1}{\sqrt{2}} \Rightarrow \frac{R^2}{L^2 \omega_c^2 + R^2} = \frac{1}{2}$$

$$\frac{1}{2} (L^2 \omega_c^2 + R^2) = R^2$$

$$L^2 \omega_c^2 = R^2 \Rightarrow \omega_c = \frac{R}{L}$$

$$3) |H(j\omega_c)| = \frac{1}{\sqrt{(RC\omega)^2 + 1}} = \frac{1}{\sqrt{2}}$$

$$R^2 C^2 \omega_c^2 + 1 = 2 \Rightarrow \omega_c = \frac{1}{RC}$$

$$4) |H(j\omega_c)| = \frac{\omega_c L}{\sqrt{R^2 + (\omega_c L)^2}} = \frac{1}{\sqrt{2}}$$

$$4) |H(j\omega_c)| = \frac{1}{\sqrt{R^2 + (\omega_c L)^2}} = \frac{1}{\sqrt{2}}$$

$$2\omega_c^2 L^2 = R^2 + \omega_c^2 L^2$$

$$\omega_c L = R \Rightarrow \omega_c = \frac{R}{L}$$

$$c) 1) |H(j\omega)| = \frac{RC\omega}{\sqrt{(RC\omega)^2 + 1}}$$

$$\omega \rightarrow \infty \text{ Hz} : |H(j\omega)| \rightarrow 1 \quad \text{HP}$$

$$\omega \rightarrow 0 \text{ Hz} : |H(j\omega)| \rightarrow 0$$

$$2) |H(j\omega)| = \frac{R}{\sqrt{(L\omega)^2 + R^2}}$$

$$\omega \rightarrow \infty \text{ Hz} : |H(j\omega)| \rightarrow 0 \quad \text{LP}$$

$$\omega \rightarrow 0 \text{ Hz} : |H(j\omega)| \rightarrow 1$$

$$3) |H(j\omega)| = \frac{1}{\sqrt{(RC\omega)^2 + 1}} \quad \text{LP}$$

$$\omega \rightarrow \infty \text{ Hz} : |H(j\omega)| \rightarrow 0$$

$$\omega \rightarrow 0 \text{ Hz} : |H(j\omega)| \rightarrow 1$$

$$4) |H(j\omega)| = \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}}$$

$$\omega \rightarrow \infty \text{ Hz} : |H(j\omega)| \rightarrow 1 \quad \text{HP}$$

$$\omega \rightarrow 0 \text{ Hz} : |H(j\omega)| \rightarrow 0$$

Problem 2

From earlier:  $\omega_c = \frac{1}{RC}$

$$\omega = 2\pi f \Rightarrow f_c = \frac{1}{2\pi RC} = 500 \text{ Hz}$$

Power loss is power consumed by resistor.

When  $f = 500 \text{ Hz}$ :

$$V_o = \frac{V_i}{\sqrt{2}} = \frac{10}{\sqrt{2}} \text{ V}$$

$$P = \frac{V^2}{R} = \frac{V_o^2}{R} = \frac{100}{2R} < 2 \Rightarrow R > 25 \Omega$$

$$C = \frac{1}{2\pi R \cdot 500} = \frac{1}{2\pi \cdot 25 \cdot 500} \Rightarrow C = 12,73 \mu\text{F}$$

The bandwidth is  $f > 500$

Time constant:  $RC = 0,3 \text{ ms}$

### Problem 3

$$\begin{aligned} \text{a) } A \cos(\omega t + \theta) &= \text{Re} \{ A e^{j\theta} e^{j\omega t} \} \\ &= \text{Re} \{ A e^{j(\omega t + \theta)} \} = \text{Re} \{ A (\cos(\omega t + \theta) + j \sin(\omega t + \theta)) \} \\ &= A \cos(\omega t + \theta) \end{aligned}$$

$$\begin{aligned} \text{b) } V_1 &= A_{V_1} e^{j\theta_{V_1}} \quad \text{and} \quad V_2 = A_{V_2} e^{j\theta_{V_2}} \\ V_1 \cdot V_2 &= A_{V_1} A_{V_2} e^{j(\theta_{V_1} + \theta_{V_2})} \end{aligned}$$

$$\begin{aligned} \text{c) } i_c &= C \frac{dV_c}{dt} \\ I_c &= C j\omega V_c \Rightarrow \frac{I_c}{V_c} = C j\omega = C\omega e^{j\frac{\pi}{2}} \\ |H(j\omega)| &= C\omega \quad \text{and} \quad \angle H(j\omega) = \frac{\pi}{2} \end{aligned}$$

The current will be the same as the

$$|H(j\omega)| = C\omega \quad \text{and} \quad \angle H(j\omega) = -\frac{\pi}{2}$$

The current will be the same as the voltage but will be amplified by  $C\omega$  and phase shifted  $\frac{\pi}{2}$ . Current leads voltage

$$d) \quad i_c = C \frac{dv_c}{dt}$$

$$\mathcal{L}\{i_c\} = \mathcal{L}\left\{C \frac{dv_c}{dt}\right\} \Rightarrow I_c(s) = Cs V_c(s)$$

$$\Rightarrow \frac{I_c(s)}{V_c(s)} = Cs$$

$$e) \quad H_1(j\omega) = H_2(s)$$

$$\Rightarrow C\omega j = Cs \Rightarrow s = j\omega$$

Comment: Only sinusoidal input

$$f) \quad \frac{dv_c(t)}{dt} + \frac{1}{RC} v_c(t) = \frac{1}{RC} v_s(t)$$

$$j\omega V_c + \frac{1}{RC} V_c = \frac{1}{RC} V_s$$

$$V_c \left( j\omega + \frac{1}{RC} \right) = \frac{1}{RC} V_s$$

$$V_c = \frac{\frac{1}{RC}}{j\omega + \frac{1}{RC}} V_s = \frac{1}{RCj\omega + 1} V_s = \frac{1 - j\omega RC}{1 + (\omega RC)^2} V_s$$

g) We know that  $v_{c,\infty}(t)$  will be  $A \cos(\omega t + \theta)$  with a new amplitude and phase

$$H(j\omega) = \frac{V_c}{V_s} = \frac{1 - j\omega RC}{1 + (\omega RC)^2}$$

$$|H(j\omega)| = \sqrt{\left(\frac{1}{1+(\omega RC)^2}\right)^2 + \left(\frac{-\omega RC}{1+(\omega RC)^2}\right)^2}$$

$$= \sqrt{\frac{1+(\omega RC)^2}{(1+(\omega RC)^2)^2}} = \sqrt{\frac{1}{1+(\omega RC)^2}}$$

$$\angle H(j\omega) = \arctan\left(\frac{\frac{-\omega RC}{1+(\omega RC)^2}}{\frac{1}{1+(\omega RC)^2}}\right) = -\arctan(\omega RC)$$

$$V_{C,\infty}(t) = \frac{A}{\sqrt{1+(\omega RC)^2}} \cos(\omega t + \theta - \arctan(\omega RC))$$

$$h) \quad sV_C(s) + \frac{1}{RC} V_C(s) = \frac{1}{RC} V_S(s)$$

$$V_C(s) \left(s + \frac{1}{RC}\right) = \frac{1}{RC} V_S(s)$$

$$V_C(s) = \frac{1}{sRC + 1} V_S(s)$$

Laplace/phaser take approx. the same amount of time

#### Problem 4

a)

$$H(s) = \frac{1}{1+Ts}$$

$$|H(j\omega)| = \frac{1}{\sqrt{1+(\omega T)^2}}$$

Low frequencies:  $|H(j\omega)| \approx 1$

High frequencies:  $|H(j\omega)| \approx 0$

$$\frac{1}{\frac{1}{T}} = \frac{1}{T} \Rightarrow \frac{1}{T}$$

$$\frac{1}{\sqrt{1+(T\omega)^2}} = \frac{1}{\sqrt{2}} \Rightarrow \omega_c = \frac{1}{T}$$

$$\omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi} \Rightarrow f_c = \frac{1}{2\pi T}$$

b) LP filter will cause a phase shift,  
so our measurements will be delayed

$$c) |H(j\omega)| = \frac{1}{\sqrt{1+(T\omega)^2}}$$

$$\omega = 2\pi f = 2000\pi$$

$$\frac{1}{\sqrt{1+(T \cdot 2000\pi)^2}} = \frac{1}{20}$$

$$T = \frac{\sqrt{399}}{2000\pi} = 3,18 \text{ ms}$$

$$f_c = \frac{1}{2\pi T} = 50,06 \text{ Hz}$$

d) We could use a higher order LP filter  
or a Butterworth filter.